Name_____ Date_____ Period_____

LESSON 2-7 · HOMEWORK PACKET · SHEET 1 OF 2

Dividing a Decimal by a Whole Number

Maryland Standard 6.NS.B.3 · Learning target – I can divide a decimal by a whole number and place the decimal point correctly.

REMEMBER

- Divide as you do with whole numbers. Put the decimal point in the quotient **straight above** the point in the dividend.
- If an amount is left over, **annex a zero** (write 0 after the last digit) and keep dividing.
- **Check:** quotient × divisor = dividend. Estimate first to catch a misplaced point.

WORKED EXAMPLE

Four friends share 22.2 kg of potting soil equally. How much does each get?

Step 1 Estimate: 22 ÷ 4 is about 5.5. Put the point straight above.

Step 2 22 ÷ 4 = 5 R2. Then 22 tenths ÷ 4 = 5 tenths R2.

Step 3 Annex a zero: 20 hundredths ÷ 4 = 5 hundredths.

Answer **5.55 kg each (5.55 × 4 = 22.2)**

PRACTICE

show your work · label units

1. Find each quotient.

a. 9.6 ÷ 3 = _____ **b.** 18.5 ÷ 5 = _____ **c.** 2.94 ÷ 7 = _____

d. 0.56 ÷ 8 = _____ **e.** 31.2 ÷ 6 = _____ **f.** 9.3 ÷ 6 = _____

2. **Bulk Rice.** A 12-pound bag of rice costs $16.80. What is the cost per pound?

Show your work.

Answer_____ per pound

3. **Relay Race.** 8 runners split a 10.2-kilometer relay equally. How far does each runner go?

Show your work.

Answer_____ km

4. **Better Buy.** Store A sells 4 cans of beans for $3.56. Store B sells 6 cans for $5.10.

a. A, per can: $ ___ **b.** B, per can: $ ___ **c.** Better buy: ____

5. **Find the Error.** Marcus wrote 4.35 ÷ 5 = 8.7. Use an estimate to explain his mistake. Then give the correct quotient.

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME At the store, find a pack of several items. Divide the price by the number of items to find the cost of one.

Name_____ Date_____ Period_____

LESSON 2-7 · HOMEWORK PACKET · SHEET 2 OF 2

Dividing by a Decimal

Maryland Standard 6.NS.B.3 · Learning target – I can divide by a decimal by making the divisor a whole number first.

REMEMBER

- Make the divisor whole: move **both** decimal points the **same** number of places right. The quotient does not change: 4.8 ÷ 0.6 = 48 ÷ 6.
- Annex zeros in the dividend if needed. Dividing by a number less than 1 gives a quotient **greater** than the dividend.

WORKED EXAMPLE

How many 0.15-liter cups can be filled from 4.5 liters of juice?

Step 1 Make 0.15 whole: 0.15 × 100 = 15.

Step 2 Do the same to 4.5: 4.5 × 100 = 450.

Step 3 450 ÷ 15 = 30. Check: 30 × 0.15 = 4.5

Answer **30 cups**

PRACTICE

show your work · label units

1. Rewrite with a whole-number divisor. Then divide.

a. 4.8 ÷ 0.6 = ____ **b.** 2.75 ÷ 0.5 = ____ **c.** 0.96 ÷ 0.12 = ____

d. 9 ÷ 0.3 = ____ **e.** 1.68 ÷ 0.04 = ____ **f.** 13.2 ÷ 1.1 = ____

2. **Pillows.** Leon has 7.5 yards of fabric. Each pillow uses 1.25 yards. How many pillows can he make?

Show your work.

Answer_____ pillows

3. **Juice Glasses.** A pitcher holds 2.4 liters of juice. How many glasses can it fill?

a. 0.3-liter glasses: _____ **b.** 0.2-liter glasses: _____

4. **Walking Path.** One lap of a park path is 0.75 mile. How many laps make each distance?

a. 4.5 miles: ____ laps **b.** 6 miles: ____ laps

5. **Find the Error.** Keisha wrote 2.4 ÷ 0.08 = 24 ÷ 8 = 3. Explain her error. Find the correct quotient.

_____

_____

TEST PRACTICE

choose the best answer

6. Which division has the same quotient as 6.72 ÷ 0.8?

A 672 ÷ 8 **B** 67.2 ÷ 8 **C** 6.72 ÷ 8 **D** 67.2 ÷ 80

7. How many 0.4-pound bags can be filled from 15.6 pounds of trail mix?

A 3.9 **B** 6.24 **C** 39 **D** 390

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Tell someone at home why 6 ÷ 0.5 is greater than 6.